



Banking Loan Data Analysis

Finance Domain

Excel+Power BI
+MySQL

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BY

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Software Utilized

Microsoft Excel

- 1.Cleaned raw dataset in Excel by removing duplicates, errors, and irrelevant fields.
- 2.Corrected data types to ensure compatibility with SQL
- 3.Formatted all date columns into YYYY-MM-DD structure for MySQL integration.
- 4.Converted the cleaned dataset into CSV format for seamless import into MySQL.
- 5.Created structured MySQL tables with appropriate data types and constraints.
- 6.Imported the cleaned CSV file into MySQL for use in queries and analysis.
- 7.Validated the migrated data to ensure accuracy, consistency, and reliability.





Software Utilized

MySQL

1. Imported the cleaned CSV file into MySQL.
 - Calculated Key Performance Indicators (KPIs) such as:
 - Total Loan Applications
 - Total Funded Amount
 - Total Amount Received
 - Average Interest Rate
 - Average Debt-to-Income Ratio (DTI)
 - Good Loan vs Bad Loan KPI's
2. Calculations to create charts for Power BI
 - Monthly Trends by Issue Date
 - Regional Analysis by State
 - Loan Term Analysis
 - Employee Length Analysis
 - Loan Purpose Breakdown
 - Home Ownership Analysis
3. Stored queries for further use in dashboards and reporting.





Software Utilized

Power Bi

Data Preparation in Power Query

- Imported data from MySQL into Power BI.
- Cleaned the dataset (removed nulls/duplicates, handled missing values).
- Checked data quality to ensure consistency.
- Changed data types (e.g., dates → Date, amounts → Decimal, IDs → Text) to make them suitable for analysis and calculations.

Calendar Table (for Time Intelligence)

- Created a Calendar Table using DAX.
- Extracted additional fields for better trend analysis:
- This ensured proper chronological sorting in visuals (charts by month, day, etc.).
-

Calculated Using DAX Measures for KPI Cards

Visualizations for Insights

- KPI Cards → Showcased total loans, funded amount, amount received, avg. interest, avg. DTI.
- Area Charts → Trends over time, applications by month, loan term, state.
- Filled (Field) Maps → Regional analysis by state.
- Stacked Bar Charts → Loan purpose, employee length, home ownership.
- Pie Charts → Good Loan % vs Bad Loan %





MySQL

KPI'S



1. Total Loan Applications.

Select count(distinct id) as
total_LoanApplication
from Bank;

total_LoanApplication
38576

1b. Total Loan Applications: for MTD.

Select count(distinct id) as
total_LoanApplication
from Bank
where Month(issue_date)=12 and
Year(issue_date)= 2021;

total_LoanApplication
4314



KPI'S



1c. Total Loan Applications for PMTD.

Select count(distinct id) as
total_LoanApplication from Bank
where Month(issue_date)=11 and
Year(issue_date)= 2021;

total_LoanApplication
4035

2.Total Funded Amount from bank.

Select sum(loan_amount) as
Total_funded_amount from bank ;

Total_funded_amount
435757075



KPI'S



2a.Total Funded Amount on MTD.

Select sum(loan_amount) as
Total_funded_amount from bank
where month(issue_date) =12;

Total_funded_amount
53981425

2b.Total Funded Amount on PMTD.

Select sum(loan_amount) as
Total_funded_amount from bank
where Month(issue_date)=11;

Total_funded_amount
47754825



3.Total Amount Received by bank.

Select sum(total_payment) as
Total_fund_recieved from Bank;

Total_fund_recieved
473070933

3a.Total Amount Received on MTD.

Select sum(total_payment) as
MTD_Total_recieved from Bank
where month(issue_date) =12;

MTD_Total_recieved
58074380



KPI'S



3b.Total Amount Received on PMTD.

```
Select sum(total_payment)
PMTD_Total_Recieved from Bank
where month(issue_date) =11;
```

PMTD_Total_Recieved
50132030

4. Average Interest Rate

```
Select sum(int_rate) as Avg_rate
from bank;
```

Avg_rate
4647.9571999999966



KPI'S



4a. Average Interest Rate on MTD.

```
Select sum(int_rate) as MTD_Avg_rate  
from bank  
where month(issue_date) =12;
```

MTD_Avg_rate
533.03959999999972

4b. Average Interest Rate on PMTD.

```
Select sum(int_rate) as PMTD_Avg_rate  
from bank  
where month(issue_date) =11;
```



KPI'S



5: Average Debt-to-Income Ratio.

Select sum(dti) as Avg_dti from Bank;

Avg_dti
5141.1906000000008

5a: Average Debt-to-Income Ratio (MTD).

Select sum(dti) as MTD_Avg_dti
from Bank

where month(issue_date) =12;

MTD_Avg_dti
589.53130000000008

5b Average Debt-to-Income Ratio (PMTD).

Select sum(dti) as PMTD_Avg_dti
from Bank

where month(issue_date) =11;

PMTD_Avg_dti
536.76530000000018



KPI'S



6 Good Loan Application Percentage

With loan_summary as(
 Select count(*) as total_loan ,
 Count(case when loan_status in ('Fully Paid','Current') Then 1
 end) as good_loan_count
from bank),
Percentage as (Select
(good_loan_count/total_loan)*100 as
Percentage_amnt from loan_summary)
Select* from percentage ;

Percentage_amnt
86.1753



KPI'S



7. Good Loan Applications

Select Count(case when loan_status in ('Fully Paid', 'Current') Then 1 end) as good_loan_application from bank;

good_loan_application
33243

8. Loan Funded Amount.

Select Sum(loan_amount) as funded_amnt from bank where loan_status in ('Fully Paid', 'Current')

funded_amnt
370224850



KPI'S



9. Good Loan Total Received Amount.

Select sum(total_payment) as
funded_amnt_recieved from bank
where loan_status in ('Fully Paid','Current');

funded_amnt_recieved
435786170

10: Bad Loan Applications.

Select Count(case when loan_status in
('Charged Off') Then 1 end) as
bad_loan_application from bank;

bad_loan_application
5333

11: Loan Funded Amount.

Select Sum(loan_amount) as funded_amnt
from bank
where loan_status ='Charged Off';

funded_amnt
65532225



KPI'S



12: Good Loan Total Received Amount.

Select sum(total_payment) as
funded_amnt_recieved from bank
where loan_status ='Charged Off';

funded_amnt_recieved
37284763

13: Bad Loan Application Percentage

With loan_summary as(
 Select count(*) as total_loan ,
 Count(case when loan_status in
('Charged Off') Then 1
 end) as bad_loan_count from bank) ,
Percentage as (Select
(bad_loan_count/total_loan)
*100 as Percentage_amnt from
loan_summary)
Select* from percentage ;

Percentage_amnt
13.8247

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KPI'S



14: Loan Status

```
Select loan_status , count(id) as
total_application ,
      Sum(loan_amount) as funded_amnt,
      Sum(total_payment) as
Recieved_payment,
      Sum(int_rate) as Avg_intRate,
      Sum(dti) as Avg_dti
from bank
group by loan_status ;
```

loan_status	total_application	funded_amnt	Recieved_payment	Avg_intRate	Avg_dti
Charged Off	5333	65532225	37284763	740.14440000000001	746.87240000000034
Fully Paid	32145	351358350	411586256	3742.0222000000016	4232.6449000000001
Current	1098	18866500	24199914	165.790600000000007	161.673299999999993



KPI'S



15: Loan Status on Month-to-date.

Select loan_status ,
Sum(loan_amount) as MTD_funded_amnt,
Sum(total_payment) as
MTD_Recieved_payment
from bank
WHERE MONTH(issue_date) = 12
GROUP BY loan_status ;

loan_status	MTD_funded_amnt	MTD_Recieved_payment
Current	3946625	4934318
Charged Off	8732775	5324211
Fully Paid	41302025	47815851



Charts



16. Monthly Trends by Issue Date.

```
select month(issue_date) As  
month_number,  
monthname(issue_date) As month_name,  
count(id) as total_application,  
sum(loan_amount) as total_funded_Amnt,  
sum(total_payment) as  
total_Amnt_Received  
from bank  
group by month_number,month_name  
order by month_number;
```

month_number	month_name	total_application	total_funded_Amnt	total_Amnt_Received
1	January	2332	25031650	27578836
2	February	2279	24647825	27717745
3	March	2627	28875700	32264400
4	April	2755	29800800	32495533
5	May	2911	31738350	33750523
6	June	3184	34161475	36164533
7	July	3366	35813900	38827220
8	August	3441	38149600	42682218
9	September	3536	40907725	43983948
10	October	3796	44893800	49399567
11	November	4035	47754825	50132030
12	December	4314	53981425	58074380



Charts



17: Loan Term Analysis

```
select term as loan_term,  
count(id) as total_application,  
sum(loan_amount) as total_funded_Amnt,  
sum(total_payment) as  
total_Amnt_Received  
from bank  
group by loan_term  
order by loan_term;
```

loan_term	total_application	total_funded_Amnt	total_Amnt_Received
36 months	28237	273041225	294709458
60 months	10339	162715850	178361475



Charts



18: Employee Length Analysis.

```
select emp_length,  
count(id) as total_application,  
sum(loan_amount) as total_funded_Amnt,  
sum(total_payment) as  
total_Amnt_Received  
from bank  
group by emp_length  
order by emp_length;
```

emp_length	total_application	total_funded_Amnt	total_Amnt_Received
< 1 year	4575	44210625	47545011
1 year	3229	32883125	35498348
10+ years	8870	116115950	125871616
2 years	4382	44967975	49206961
3 years	4088	43937850	47551832
4 years	3428	37600375	40964850
5 years	3273	36973625	40397571
6 years	2228	25612650	27908658
7 years	1772	20811725	22584136
8 years	1476	17558950	19025777
9 years	1255	15084225	16516173



Charts



19: Loan Purpose Breakdown

```
select purpose,  
count(id) as total_application,  
sum(loan_amount) as total_funded_Amnt,  
sum(total_payment) as  
total_Amnt_Received  
from bank  
group by purpose  
order by purpose;
```

purpose	total_application	total_funded_Amnt	total_Amnt_Received
car	1497	10223575	11324914
credit card	4998	58885175	65214084
Debt consolidation	18214	232459675	253801871
educational	315	2161650	2248380
home improvement	2876	33350775	36380930
house	366	4824925	5185538
major purchase	2110	17251600	18676927
medical	667	5533225	5851372
moving	559	3748125	3999899
other	3824	31155750	33289676
renewable_energy	94	845750	898931
small business	1776	24123100	23814817
vacation	352	1967950	2116738
wedding	928	9225800	10266856



Charts



20: Home Ownership Analysis

```
select home_ownership,  
count(id) as total_application,  
sum(loan_amount) as total_funded_Amnt,  
sum(total_payment) as  
total_Amnt_Received  
from bank  
group by home_ownership  
order by home_ownership;
```

home_ownership	total_application	total_funded_Amnt	total_Amnt_Received
MORTGAGE	17198	219329150	238474438
NONE	3	16800	19053
OTHER	98	1044975	1025257
OWN	2838	29597675	31729129
RENT	18439	185768475	201823056



Charts



21: Regional Analysis by State.

```
select address_state as state,  
count(id) as total_application,  
sum(loan_amount) as total_funded_Amnt,  
sum(total_payment) as  
total_Amnt_Received  
from bank  
group by state  
order by state;
```

state	total_application	total_funded_Amnt	total_Amnt_Received
AK	78	1031800	1108570
AL	432	4949225	5492272
AR	236	2529700	2777875
AZ	833	9206000	10041986
CA	6894	78484125	83901234
CO	770	8976000	9845810
CT	730	8435575	9357612
DC	214	2652350	2921854
DE	110	1138100	1269136
FL	2773	30046125	31601905
GA	1355	15480325	16728040
HI	170	1850525	2080184
IA	5	56450	64482
ID	6	59750	65329
IL	1486	17124225	18875941
IN	9	86225	85521
KS	260	2872325	3247394





Power BI

1. Making a Calender table in Dax .

- `Calender=CALENDAR(min(Bank[issue_date]),max(Bank[issue_date]))`

2. Extracting Month num , Day num , Day, formatting day name using new column .

- `Month_number = Month(Calender[Date])`
- `day_numb = WEEKDAY(Calender[Date],1)`
- `Day = day(Calender[Date])`
- `day_name=`
`Format(Calender[day_numb],"ddd")`



3. Calculating all Kpi's for total Loan Application.

- **Total Loan Application** =
`DISTINCTCOUNT(Bank[id])`
- **MTD TLA** = `TOTALMTD([Total Loan Application],Calender[Date])`
- **PMTD TLA** = `CALCULATE([Total Loan Application],PREVIOUSMONTH(DATESMTD(Calender[Date])))`
- **MoM TLA** = `DIVIDE(Calender[MTD TLA],Calender[PMTD TLA])-1`



4. Calculating all Kpi's for Total Amount Received

- **Total Amount Received** =
 $\text{Sum}(\text{Bank}[\text{total_payment}])$
- **MTD TAR** = $\text{TOTALMTD}([\text{Total Amount Received}], \text{Calender}[\text{Date}])$
- **PMTD TAR** = $\text{CALCULATE}([\text{Total Amount Received}], \text{PREVIOUSMONTH}(\text{DATESMTD}(\text{Calender}[\text{Date}])))$
- **MoM TAR** = $\text{DIVIDE}(\text{Calender}[\text{MTD TAR}], \text{Calender}[\text{PMTD TAR}]) - 1$



5. Calculating all Kpi's for Average Interest Rate.

- **AVG INR RATE** = Average(Bank[int_rate])
- **AVG MTD** = TOTALMTD([AVG INR RATE],Calendar[Date])
- **AVG PMTD** = CALCULATE([AVG INR RATE],PREVIOUSMONTH(DATESMTD(Calendar[Date])))
- **AVG PMTD** = CALCULATE([AVG INR RATE],PREVIOUSMONTH(DATESMTD(Calendar[Date])))



6. Calculating all Kpi's for Average DTI Rate.

- **Avg DTI** = `Average(Bank[dti])`
- **AVG MTD DTI** = `TOTALMTD([Avg DTI],Calender[Date])`
- **AVG PMTD DTI** = `CALCULATE([Avg DTI],PREVIOUSMONTH(DATESMTD(Calender[Date])))`
- **MoM AVG DTI** = `DIVIDE(Calender[AVG MTD DTI],Calender[AVG PMTD DTI])-1`



7. Calculating all Kpi's for Good Loans.

- **Good Loan Application** =
`CALCULATE(COUNTA(Bank[loan_status]),Bank[loan_status]="Fully Paid" || Bank[loan_status] = "Current")`
- **Good Loan Percentage** = `DIVIDE([Good Loan Application],[Total Loan Application],0)`
- **Good Loan Funded Amount** =
`CALCULATE (Calender[Total Funded Amount], Bank[loan_status] IN { "Fully Paid", "Current" })`
- **Good Loan Funded Amount** =
`CALCULATE ( Calender[Total Funded Amount], Bank[loan_status] IN { "Fully Paid", "Current" })`



8. Calculating all Kpi's for Bad Loans.

- **Bad Loan Application** =
`CALCULATE(COUNTA(Bank[loan_status]),Bank[loan_status]="Charged Off")`
- **Bad Loan Percentage** = `DIVIDE([Bad Loan Application],[Total Loan Application],0)`
- **Bad Loan Funded Amount** =
`CALCULATE (Calender[Total Funded Amount], Bank[loan_status] IN {"Charged off" })`
- **Bad Loan Amount Recieved** =
`CALCULATE ( Calender[Total Amount Received], Bank[loan_status] IN {"Charged Off" })`



Loan Portfolio Analysis | Overview



Overview

Summary

Details

STATES

All



GRADE

All



PURPOSE

All



Total Loan Application

MTD 39K MoM 6.9%
4314

Total Funded Application

54M 13%
\$436M

Total Amount Received

MTD 58M MoM 15.8%
\$473M

Average Interest Rate

MTD 0.12 MoM 3.5%
12.05%

Average Dti

MTD 13.7% MoM 2.7%
13.3%

Good Loan Issued



Good Loan Application

33K

Good Loan Funded Amount

\$370.2M

Good Loan Amount Recieved

\$435.8M

Bad Loan Issued



Bad Loan Application

5K

Bad Loan Funded Amount

\$65.5M

Bad Loan Amount Recieved

\$37.3M

Loan Status	Loan Application	Funded Amount	Amount Received	Average Interest	Average Dti
Fully Paid	32145	\$351,358,350	\$411,586,256	11.64%	13.2%
Charged Off	5333	\$65,532,225	\$37,284,763	13.88%	14.0%
Current	1098	\$18,866,500	\$24,199,914	15.10%	14.7%
Total	38576	\$435,757,075	\$473,070,933	12.05%	13.3%

Loan Portfolio Analysis| Details



Overview

Summary

Details

STATES

All

GRADE

All

Good Vs Bad Loan

All

Total Loan Application

MTD 39K MoM 6.9%
4314

Total Funded Application

MTD \$436M MoM 13%
54M

Total Amount Received

MTD \$473M MoM 15.8%
58M

Average Interest Rate

MTD 12.05% MoM 3.5%
0.12

Average Dti

MTD 13.3% MoM 2.7%
13.7%

Id	Purpose	Home ownership	Grade	sub_grade	Issue Date	Funded Amount	Interest Rate	Installment	Amount Received
802401	Debt consolidation	RENT	F	F1	July 11, 2021	\$35,000	20.25%	\$1,305.19	\$23,396
737345	Debt consolidation	OWN	G	G1	April 11, 2021	\$35,000	20.11%	\$1,302.69	\$6,457
741236	Debt consolidation	MORTGAGE	E	E5	May 11, 2021	\$35,000	19.69%	\$1,295.21	\$39,727
810107	Debt consolidation	RENT	E	E4	July 11, 2021	\$35,000	19.29%	\$1,288.1	\$27,049
872736	Debt consolidation	MORTGAGE	E	E4	September 11, 2021	\$35,000	19.29%	\$1,288.1	\$45,955
1038229	Debt consolidation	RENT	E	E2	December 11, 2021	\$35,000	19.03%	\$1,283.5	\$46,206
879702	Debt consolidation	RENT	E	E1	September 11, 2021	\$35,000	18.64%	\$1,276.6	\$45,958
936915	Debt consolidation	RENT	E	E1	October 11, 2021	\$35,000	18.64%	\$1,276.6	\$37,414
1025160	Debt consolidation	MORTGAGE	E	E1	November 11, 2021	\$35,000	18.64%	\$1,276.6	\$45,279
755241	credit card	RENT	E	E2	May 11, 2021	\$35,000	18.39%	\$1,272.2	\$44,791
971188	Debt consolidation	MORTGAGE	D	D5	October 11, 2021	\$35,000	18.25%	\$1,269.73	\$45,710
988119	Debt consolidation	MORTGAGE	D	D5	November 11, 2021	\$35,000	18.25%	\$1,269.73	\$36,545
991612	credit card	RENT	D	D5	October 11, 2021	\$35,000	18.25%	\$1,269.73	\$36,864
995368	Debt consolidation	RENT	D	D5	October 11, 2021	\$35,000	18.25%	\$1,269.73	\$42,529
796251	Debt consolidation	OWN	E	E1	July 11, 2021	\$35,000	17.99%	\$1,265.16	\$45,546
700298	small business	MORTGAGE	E	E5	March 11, 2021	\$35,000	17.88%	\$1,263.23	\$45,476

Loan Portfolio Analysis | Summary



Overview

Summary

Details

MENU

Total Funded Amount

STATES

All

GRADE

All

Good Vs Bad Loan

All

Total Loan Application

MTD 39K
4314 MoM 6.9%

Total Funded Application

MTD \$436M
54M MoM 13%

Total Amount Received

MTD \$473M
58M MoM 15.8%

Average Interest Rate

MTD 12.05%
0.12 MoM 3.5%

Average Dti

MTD 13.3%
13.7% MoM 2.7%

Loan Application by Month



Application by State



Loan Application by term

term

36 mon...
60 mon...



Loan Application by Purpose



Loan Application by Emp_length



Loan Application by Home Ownership

RENT

18K

MORTGAGE

17K

The End